

DISTRIBUTION OF HEAVY MINERALS IN INDIAN BEACH SAND DEPOSITS

India is gifted with valuable resources of beach sand minerals. Indian coast-line along Kerala, Tamil Nadu, Odisha, and Andhra Pradesh where significant deposits of different minerals are available is presented below. The important economic minerals are such as -

Ilmanite ($\text{FeO} \cdot \text{TiO}_2$)

Colour: Black / steel grey

Lustre: Metallic

Composition: Iron + Titanium oxide



Rutile (TiO_2)

Colour: Reddish brown / yellow / black

Crystal form: Needle-like or prismatic

Composition: Pure titanium dioxide



Monazite Sand (India, Kerala)

Monazite ($\text{Ce,La,Y,Th} \text{PO}_4$)

Colour: Yellow to brown

Composition: Rare earth phosphate (contains thorium)

Occurrence: Beach placer sands



The elements in the parentheses are the "ingredients" that can swap places within the mineral's crystal structure.

- **Ce (Cerium):** Usually the most abundant element in monazite.
- **La (Lanthanum):** A major rare earth element used in glass manufacturing and batteries.
- **Y (Yttrium):** Used in LEDs and superconductors.
- **Th (Thorium):** A radioactive element. Because monazite contains thorium, it is naturally radioactive, which is why it's often studied as a potential fuel for nuclear reactors.

The PO₄ at the end tells us that monazite is a phosphate mineral. This is the "backbone" that holds the rare earth metal atoms together.

Garnet

Appears as reddish to dark brown grains

Mixed with other heavy minerals like:

Ilmenite (black)

Rutile (brown)

Zircon (light colored)

Usually seen as small rounded or angular particles



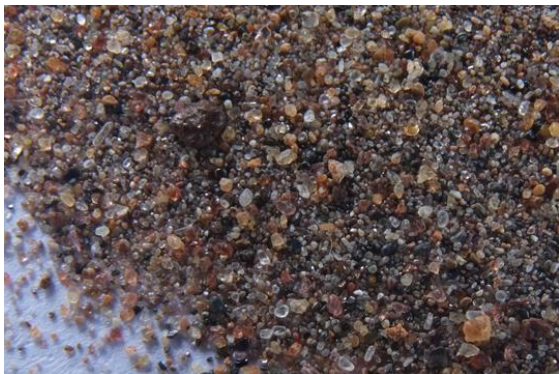
Sillimanite ($\text{Al}_2\text{O}_3 \cdot \text{SiO}_2$)

Colour: Light brown / grey / whitish

Shape: Often elongated or fibrous grains

Texture: May appear needle-like or columnar

Less shiny compared to ilmenite or rutile



Zircon ($\text{ZrO}_2 \cdot \text{SiO}_2$)

Colour: white, yellow, or light brown

Appearance: Shiny, glassy (vitreous lustre)

Shape: Small rounded or prismatic grains

Looks lighter compared to black minerals like ilmenite

Varying in size, concentration and grade ranging from 5% to 45%.



Uses of different products

Ilmenite & Rutile:

As Titanium (TiO₂) white pigment: Used in Paints/ Varnishes, Plastic, Paper, Rubber, Printing ink, coated fabric textiles, Cosmetics, sun protection creams and Pharmaceuticals.

As Titanium (Ti) sponge/metal: Used in Chemical industry, Aerospace & Aviation industry, Surgical equipments, Electrical turbines tubing, Bullet proof vests, Different alloys in Iron & steel industry, Immersion heater tubes, Consumer goods, Spectacle frames, Golf clubs.

Used for coating of welding electrodes.

Zircon:

Used in Ceramics, Foundries, Refractories, Glazing tiles, Television and Computer monitors.

It is also used in manufacture of Zirconium chemicals/metal, American diamond, Scratch free bracelets, Cutting tools, Ytria Zirconia as oxygen sensors. Zircon free from Hafnium is used in nuclear reactors as cladding tubes to hold nuclear fuel.

Sillimanite:

Mainly used for the manufacture of High grade refractory bricks, high Alumina Refractories, Cement kilns and Heat treatment furnaces.

Garnet:

Used for manufacture of Blasting media, Abrasives, Grinding wheels, Mosaic cutting stones, Decorative wall plasters, Ceramics, Polishing of picture tubes, Glass polishing & Antiskid surface for roads, air strips, runways, water filter, water jet cutting,

Artificial Granite tiles/Heavy duty floor tiles, cleaning of casings/pipes in petroleum industry and as a gemstone.

Monazite:

Extraction of thorium concentrate and rare earth compounds.

Rare earth chlorides: Widely used in the manufacture of Misch metal used for lighter flints, for the production of catalysts for cracking, for the manufacture of metallic soaps which is used as Dryer in paints, starting material for the production of pure rare earths & rare earth compounds, removal of organic impurities and decolourisation of paper mills effluents and for the manufacture of special ferrous casting.

Rare earth Fluorides: Used in manufacture of arc carbon electrodes to increase the arc intensity, rare earth alloys, for production of nodular cast iron, special steels.

Rare earth Oxides: In the arc carbon industries to increase the arc intensity, the emitted light is identical to natural sun light, for glass polishing in Optical.

Cerium oxide: Used in Polishing optical lenses, plate glasses, TV tube face plates; prism etc. finds application in semiconductor devices, UV absorber in Glasses, radiation protective glasses, yellow color and decolorizing etc.

Cerium hydrates: Used in manufacture of polishing composition, in the decolorizing of glass, as an opacifier.

Didymium compound: Use in manufacture of pure rare earth compound, in glass, ceramics, and nuclear & electronic industries and for improving the workability of stainless steel alloys.

Samarium oxide: For the extraction of samarium metal, high-performance Samarium-Cobalt (SmCo) magnets, which operate at high temperatures in aerospace and defense applications.

Samarium cobalt; used as permanent magnet with high coercive force and high magnetic energy.

Gadolinium oxide: In the manufacture of Gadolinium-Gallium-Garnet (GGG) substrates of magnetic bubbles and Microwave garnets.

Yttrium oxide: In the manufacture of phosphor for color tubes and Fluorescents tubes, super conductors and artificial gems.

Europium oxide: As an activator in preparing phosphors for color TV tubes & fluorescent lamps.

Europium: Three bond lamps, cathode ray tubes (CRT), flat plasma TV screen and phosphors.

Terbium: Luminescence, phosphors.

Dysprosium: Luminescence, phosphors, application for nuclear industry, ceramics.

Holmium: Application for nuclear industry, ceramics, laser.

Erbium: Nuclear reactors, ceramics, glass coloring, optic fibers, medical applications, laser.

Thorium Nitrate: Gas handling Industry

Thorium Oxide: Fluorescence tubes & starters; Catalyst for Petroleum industry

Uranium Oxide: Nuclear Industry

Tri sodium phosphate: Descaling, Degreasing, Detergents.

FACTORS CONTROLLING FORMATION OF BEACH SAND DEPOSITS

Heavy mineral sands are a class of ore deposit which is an important source of rare earth elements and the industrial minerals such as diamond, sapphire, garnet, and other precious metals or gemstones. Placer deposits are formed usually in beach environments by concentration due to the specific gravity of the mineral grains. Heavy minerals (aside from gold placers) exist within streambeds, but most are relatively small and of a low grade. The grade of a heavy mineral sand ore deposit is usually low.

The cut-off grades of heavy minerals, as a **total heavy mineral (THM)** concentrate from the raw sand, in most ore deposits of this type is around 1% heavy minerals, although several are of higher grade. Total heavy mineral concentrate (THM), components are typically for:

Zircon- 1% to 50% of THM,

Ilmenite- 10% to 60% of THM

Rutile- 5% to 25% of THM

Leucoxene- 1% to 10% of THM

Typically magnetite, Chromite, garnet which is trash minerals usually accounts the remaining bulk of the THM content.

The Beach sand minerals are most recently originated minerals deposits. These are originated in southern hemisphere due to continental drifts. Due to motion between different plates and repeated weathering i.e. heating and cooling, cracks are developed in between the rocks and water gets entered into these cracks. Parent rock is subjected to weathering and erosion and transferred along with sea water and finally sediment in a suitable basin. Continued erosion due to air and water takes place and concentration goes on increasing with time.

The source of heavy mineral sands is within the erosional areas of a river where the eroded minerals are dumped into the ocean, thereafter the sediments are caught up in littoral or long shore drift. Rocks are sometimes eroded directly by the wave action and washed up onto beaches and the lighter minerals are sort through.

The source rocks determine the composition of the economic minerals. **Usually granite is the source of zircon, Rutile, monazite, and some Ilmenite. The source of Ilmenite, garnet is ultramafic and mafic rocks, such as kimberlite or basalt.** Garnet is sourced commonly from metamorphic rocks, **such as amphibolite schists. Precious metals are generally sourced from deposits hosted within metamorphic rocks.**

MINES ACT, RULES & REGULATIONS RELATING TO BEACH SAND MINING IN INDIA

Acts, Rules & Regulation applicable for beach sand mining:-

Mines Act, 1952

Mines and Minerals (Development and Regulation) Act, 1957 MMDR Act, 1957

Atomic Energy Act, 1962

Environment Protection Act, 1986

Forest (Conservation) Act, 1980

Mines Rules, 1955

Mines Crèche Rules, 1966

Mines Vocational Training Rules, 1966

Mineral Concession Rules, 1960

Mineral Conservation & Development Rules, 1988

Indian Electricity Rules, 1956

Odisha Mineral Rules, 2007

Metalliferous Mines Regulations, 1961

These mines are governed by the following Regulatory Bodies:

1. Director General of Mines Safety, Dhanbad
2. Indian Bureau of Mines, Nagpur
3. Director of Mines, Orissa

Acts, Rules and Regulations regulated / enforced by the Director General of Mines Safety are:

1. Mines Act, 1952
2. Metalliferous Mines Regulations, 1961
3. Mines Rules, 1955
4. Mine Vocational Training Rules, 1966
5. Indian Electricity Rules, 1956

Royalty:

Section 9 of MMDR Act 1957 fixed for a period 3 years

Ilmenite/Rutile/Zircon: 2% of sale price on Ad valorem basis. “Ad valorem” is a Latin term meaning:

“according to value”

So, the charge depends on the value (price) of the mineral, not just quantity.

Sillimanite: 2.5% of sale price on Ad valorem basis.

Garnet: 3% of sale price on Ad valorem basis.

Monazite: Rs125/tonne (Ad-valorem not applicable)

For non-atomic minerals (Garnet & Sillimanite), sale price = IBM published value + 20%

For atomic minerals (Ilmenite, Rutile, Zircon), sale price = invoice price in case of domestic sale

= FOB or CIF price less handling expenses in case of export

CIF – Cost, Insurance, and Freight up to destination port, Everything in FOB plus: Pays shipping (freight), Pays insurance

Dead rent:

In case of natural calamity, if there is stoppage of production and also there is no stock for sales, dead rent will be paid.

Environmental Protection Act, 1986:

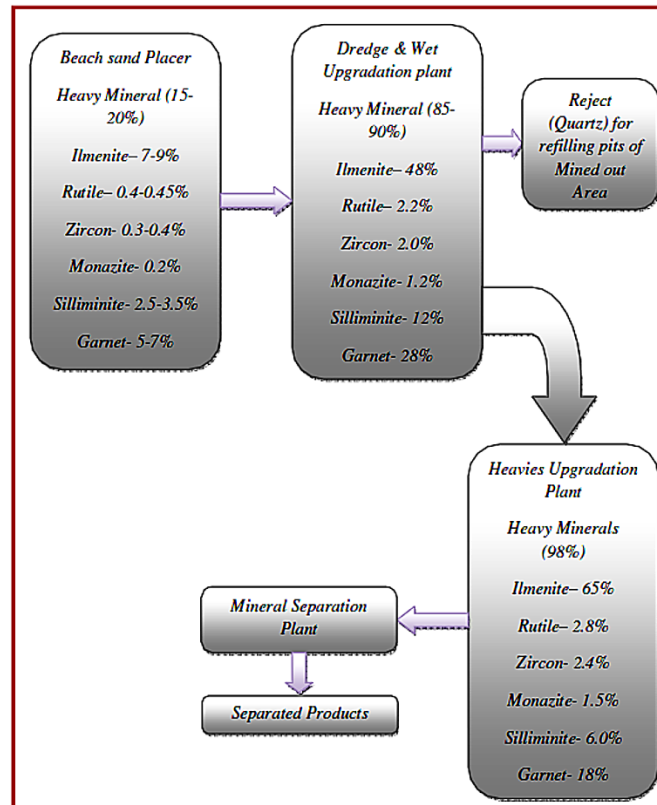
It is also known as Umbrella Act, 1986. Coastal Regulation Zone (CRZ), February, 1991: It is measured 500 m from the appropriate base line towards the land side and 200 m measured towards the sea from high tide line.



Basic Unit Operations

The complete operation at Indian Rare Earths Ltd can be broadly classified as:

1. Mining/Dredging & Preconcentration [Dredging & Wet Upgradation Plant (DWUP)]
2. Heavy mineral Upgradation [Heavy Mineral Upgradation Plant (HUP)]
3. Mineral separation [Mineral Separation Plant (MSP)]



Dredging and Wet Upgradation Plant (DWUP)

The OSCOM have a huge placer deposit with a length of 18 Kms and width of 1.5 Kms running parallel to the coast of Bay of Bengal. Northern boundary of the mining area is Rusikulya River and southern boundary is Gopalpur town. The estimated reserves are 230 MT (1.5m below MSL) and 440 MT (6m below MSL). Mining is divided in two sections; North section and South section. The whole mining area is classified in three zones; plenty zone, Intermediate zone and rare zone.

This plant is in complete floating condition. It floats in a pond having diameter 200m and depth of 6m. The whole unit can be divided in two groups:

- 1) Mining/dredging
- 2) Pre-concentration

Mining:

The mining procedure followed for the excavation of beach sand is completely different from the surface or underground mining. The procedure followed is commonly known as DREDGING.

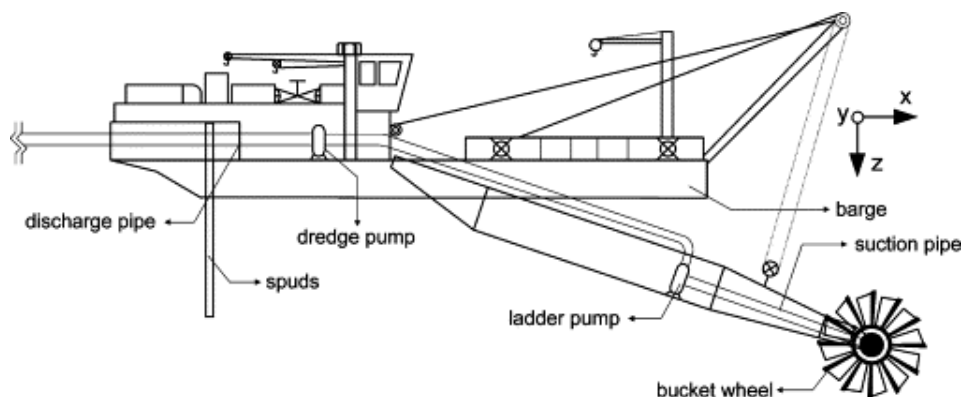
Dredging can be defined as an excavation activity or operation which is carried out at partly underwater, in shallow depth of sea or fresh water areas with the purpose of gathering up bottom sediments and disposing of them at a different location. A dredge is a device used for dredging for scraping or sucking the seabed. A dredger is a boat or ship equipped with a dredge. The process of dredging creates wastes (excess material), which are conveyed to another location different from the dredged area. The dredge Capacity is about 500 TPH and dredge

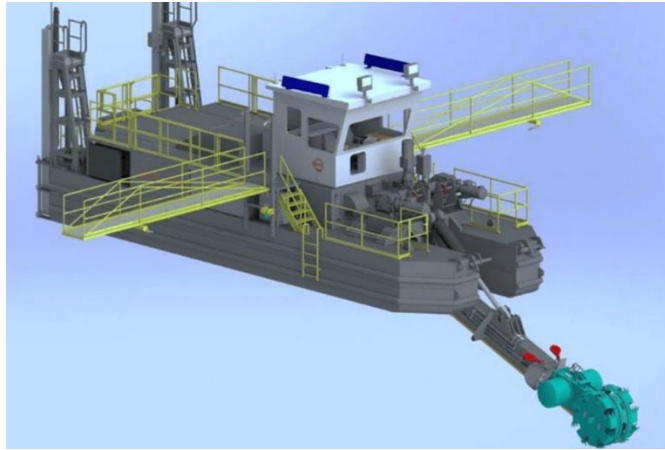
depth is nearly 6m. The RPM of the motor used is 368 with a power of 525KW. In OSCOM, There are two different types of dredge used for dredging purposes:

Cutter Suction Dredge: A cutter-suction dredger's (CSD) suction tube has a cutter head at the suction inlet, used to loosen the earth and convey it to the suction mouth. The cutter can also be used for hard surface materials like gravels or rocks. The dredged material is usually sucked up by a wear-resistant centrifugal pump which is discharged through a pipe line or to a barge. In recent years, in order to excavate harder rock without blasting, dredgers with more powerful cutters have been built.



Bucket wheel Dredge: The bucket-wheel dredge is identical to the cutter suction dredge apart from the position of wheel excavator which is used in lieu of the rotary cutter. In both the cases, the cutter attached to the ladder rotates and cuts the sand and thus make a suspension of sand in water which is pumped out to the Trommel.

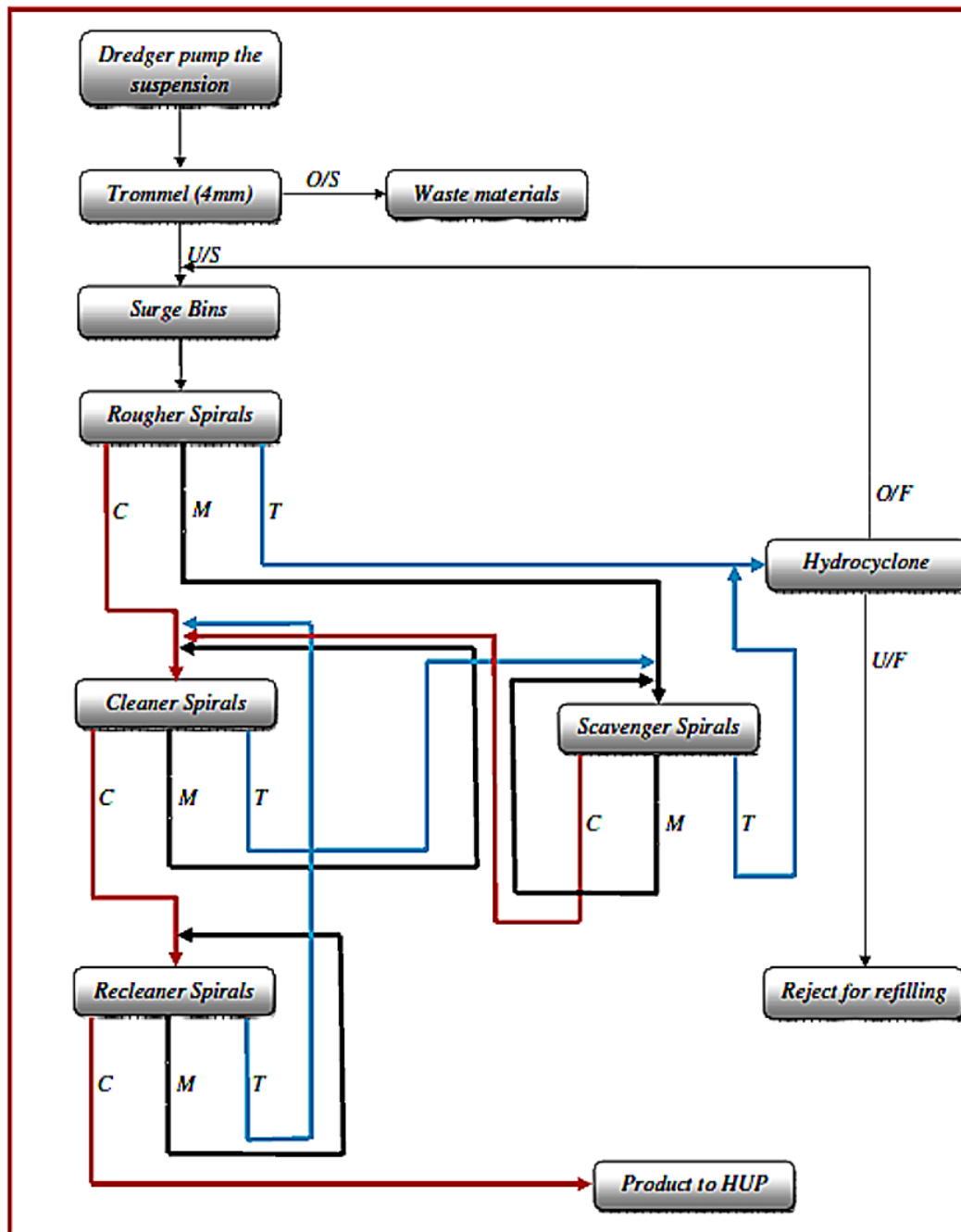




Dredger at OSCOM

Pre-concentration

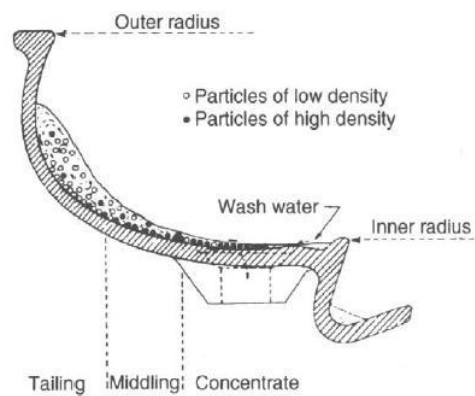
In this stage, the amount of heavies is upgraded up to 90%. Then it is sent to HUP for further treatment. The suspension formed by the dredger is pumped to the Trommel with aperture size is 4mm, which rotates at 6RPM with a motor power of 37KW. The undersize of the Trommel is sent to bins from where it is sent to the spirals for concentration and the oversize usually containing pebbles, grass and other waste materials. In spirals 4-stage of cleaning takes place i.e. Rougher, Cleaner, Recleaner and Scavenger. Concentrate from the rougher spirals is sent to cleaner spirals, middlings to Scavenger spirals and tailings to the Hydrocyclone from where the overflow (water) is sent to Trommel discharge and the underflow is the rejects (sand) which is thrown for back filling of the pits. Cleaner concentrate is sent to Recleaner, middlings is recirculated and tailings to the scavenger. Recleaner concentrate is final concentrate which is sent to HUP (Heavy upgradation plant), middlings is recirculated and tailings are sent to cleaner. The scavenger concentrate is sent to cleaner circuit, middlings is recirculated and tailings are sent to Hydrocyclone. By this process the amount of heavies are improved to 90-92%.



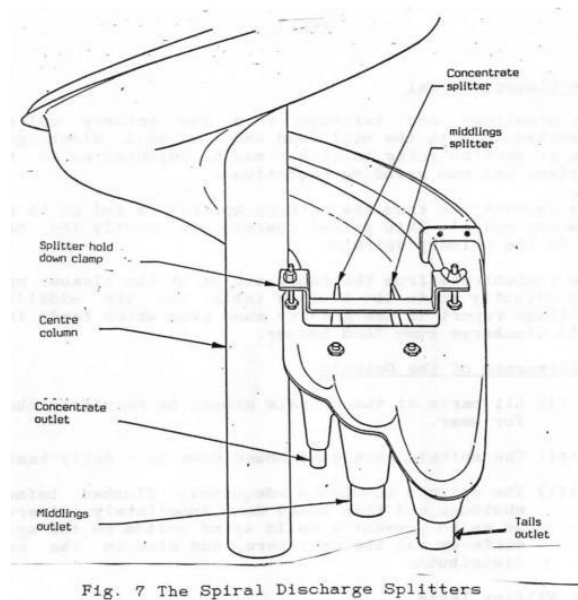
US: under size, OS: Over size, C: Concentrate, M: middling, T: tailings

Spirals

Spirals are gravity separators where slurry is fed to at a pulp density of about 20-25%, the size range is commonly 3mm to 75 microns and as the slurry flows down the curved channel, lighter particles due to action of centrifugal forces will report to the outer area of the spirals as tails while the heavier particles are pulled inward due to drag force and report to the inner area of the spiral as concentrate, thus effecting the separation. Modern spirals are constructed from fiberglass and plastic and can treat 1-3TPH of feed effectively. The operating parameters are feed rate, pulp density, feed grade, splitter openings, position of distributors, diameter of spirals etc.



PHOTOGRAPHS OF SPIRALS AND PRINCIPLE Image



Heavy Upgradation Plant (HUP)

This is the stage where the concentration of heavy is further upgraded to 98%. The feed to HUP plant is the Recleaner concentrate of the DWUP where heavy minerals concentration is 90-

92%. The heavy minerals are received from Dredge and Wet upgradation plant (DWUP) at the Re-pulping area of Mineral Separation Plant (MSP). These materials are pumped for upgradation in heavy Upgradation Plant (HUP) at MSP. In HUP, the feed is treated in gravity separation equipment like spirals and Hydrocyclones and the unwanted lighter material is pumped to reject dumping yard and the upgraded heavy mineral is pumped to the dry feed yard through Hydrocyclones for natural dewatering and finally it is fed to main plant for drying and further separation.

MINERAL SPECIFICATION AT HUP

Minerals	HUP Output Minerals Percentage	HUP Reject Quality
Ilmenite	65 – 68%	6 – 7%
Rutile	2.6 – 2.8%	0.6 – 0.7%
Zircon	2.5 – 2.7%	0.4%
Monazite	1.5%	0.05%
Sillimanite	6.5 – 7%	22 – 24%
Garnet	15 – 17%	39 – 40%
Quartz	2 – 3%	25 – 27%
Other	0.5 – 0.6%	2 – 3%

Mineral Separation Plant (MSP)

In the mineral separation plant (MSP), the heavy minerals like Ilmenite, Rutile, monazite, zircon and Silliminite are separated from the upgraded feed minerals on the basis of their physical properties like electrostatic & magnetic property, surface characteristics and specific gravity. The plant comprised of different floor and equipped with different material handling equipment like bucket elevators, belt conveyors, screw conveyors, drag conveyors to facilitate smooth transport of materials to the desired machines for example Rotary Dryer, High tension separators, Magnetic separators, Shaft dryers, Electrostatic Separators etc. Beside this, there is a wet processing circuit which comprises of Spirals, Floatex, Wet Tables, Flotation cells etc and Slurry Pumps are used to transport materials from one point to another.

Annual Production Capacity of MSP

Ilmenite: 220000 TPA

Rutile: 7400 TPA

Zircon: 5000 TPA

Sillimanite: 8000 TPA

Monazite: 2350 TPA

Obstacles of placer mining

Land Acquisition is an issue. The area under lease has decreased owing to various social & environmental issues.

These minerals are strategic assets. Secrecy is maintained according to shipment and production data.

The CRZ notification has further divested IREL for carrying out dredging activities within 100 meters of the High Tide Line.

Fresh water availability for the artificial pond is minimized.

Dredging creates disturbances to the aquatic ecosystems. Dredge spoils may contain toxic chemicals which have an adverse impact on the disposal area; The process of dredging often dislodges chemicals residing into the water column.

The following impacts to the environment due to the activity of dredging :

- Release of toxic chemicals like heavy metals
- Increase in turbidity, which affects aquatic species metabolism
- Secondary effects from water column contamination, heavy metals, other persistent organic toxins, via food chain uptake and concentrations of these toxins in higher organisms including humans.
- Difficulty in marsh productivity from sedimentation
- Impacts to avifauna (All types of birds in a region) as they have to prey upon contaminated aquatic organisms
- Secondary impacts to the aquatic and benthic organisms' [Organisms living at the bottom of water bodies (sea, river, lake) Examples: worms, crabs, molluscs, bacteria] mortality and metabolism.

ADVANTAGES OF PLACER MINING

Processing and transport cost is reduced up to 40-50% as these are loose materials.

Safety issues are not major as compared to underground mining.

The deposits in Orissa were very old; however in Kerala and Kanyakumari, still the deposits are being readily formed and extracted by beach washing.

Produces pure quartz based sand which is dumped, hence, the waste dump does not contain other heavy minerals